

Fairness-Aware Classification over Incomplete Data

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Abstract

The missing values widely existed in tabular data hinder the effective analysis of algorithmic fairness. Existing fairness intervention algorithms incorporate constraints or regularizers to reduce discrimination which rely on the complete information. They cannot effectively handle common tabular data with missing values in both *sensitive* and *non-sensitive attributes* without imputation. In this paper, we propose a *novel Transformer-based fairness-aware prediction model FATE* that mitigates the bias introduced by missing values to achieve algorithmic fairness without imputation. FATE consists of two modules, i.e., **an incomplete data encoding (IDE) module and a debiased representation learning (DRL) module. IDE designs an incomplete tabular data embedding strategy and a missingness-aware Transformer block to effectively learn the observed data distribution and the missing state information.** DRL converts fairness into attention parity when the sensitive attributes are completely missing. It offers a debiased attention mechanism to normalize attention weights in the attention score calculation process. We theoretically prove that, the differences in attention scores can represent the demographic disparities among sensitive groups which in FATE are bounded by a constant, substantially minimizing the group discrimination. Extensive experiments on three public real-world datasets demonstrate that, FATE, with the *competitive* fairness, yields more than 22% accuracy gain, compared to the state of the arts.

CCS Concepts

• **Information systems** → **Information integration; Data mining.**

Keywords

Missing data, Algorithmic fairness, Incomplete data

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1 Introduction

With the growing influence of algorithmic decision-making systems on many real-life applications such as job employment [33], loan approvals [26], personalized recommendation [58], and crime assessments [4], concerns about *fairness* have intensified. Mounting evidence has suggested that, powerful machine learning (ML) algorithms can be unfair and lead to disproportionately unfavorable decisions for marginalized groups [4, 30].

Incomplete data are widespread, presenting notable challenges in real-world analysis. Due to various reasons like improper data collection [46, 48, 49], limited resources [34, 51], and privacy concerns [39], the missing distribution of the tabular data is complex and varies greatly over applications. Incomplete data can contain sensitive attributes missing and non-sensitive attributes missing. In many cases with privacy protection [30], ethical considerations [55], and regulatory restrictions [25], it is usually unfeasible to collect and use *sensitive* attributes (e.g., race, sex, and age). Hence, sensitive attributes might be completely absent in data analysis.

Missing data negatively impacts algorithmic fairness [24, 56]. Data incompleteness has been identified as a critical source of algorithmic bias across different domains. In recommender systems, lack of historical data from underrepresented groups biases recommendations toward popular items, reinforcing long-tail unfairness. Similarly, in healthcare applications, systematic missingness of data for certain demographic groups creates representation gaps that significantly degrade model performance for those underrepresented populations. This differential data quality across groups not only reduces predictive accuracy but also contributes to biased algorithmic decisions [14, 36]. As a consequence, it is essential to promote fairness of prediction over incomplete data, especially with the absence of sensitive attributes.

Existing fairness-aware prediction algorithms ignore cases where *both* non-sensitive and sensitive attributes are missing. In particular, traditional fairness intervention algorithms over *full knowledge* of sensitive attributes, using embedding constraints or regularizers [1, 20, 37], do not consider the missing non-sensitive attributes. One

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exceptional method [24] (that considers incomplete non-sensitive attributes) has the assumption of *fully observed* sensitive attributes. It defines the fairness based on sensitive attributes, and thus does not work when sensitive attributes are completely missing.

The closest related algorithms [25, 53] are the ones designated to tackle the fairness over the data with (fully or partially) missing sensitive attributes. All of them assume the full completeness of non-sensitive attributes. They cannot directly support the case where both sensitive and non-sensitive attributes are missing. Imputing missing non-sensitive attributes, as a pre-processing step before fairness intervention, could be a possible solution. However, the fact is that, almost all of imputation methods primarily operate under the ideal assumption of *missing completely at random* (MCAR). When dealing with real-world complex incomplete data, such as in most recommender systems [19, 40], rating data often exhibits selection bias, resulting in *missing at random* (MAR) and *missing not at random* (MNAR). As a result, using the imputation-then-classification approach often leads to additional bias, significantly impacting prediction accuracy and fairness.

Therefore, we are faced up with two major challenges to effectively make fairness-aware predictions over incomplete data. *First, how to fully utilize the information contained in the missingness while minimizing or preventing the imputation bias, so as to enhance prediction accuracy (CH1)?* *Second, how to effectively evaluate and optimize fairness when sensitive attributes are absent (CH2)?*

In this paper, we propose a Transformer-based fairness-aware prediction model FATE to enhance both algorithmic accuracy and fairness in the presence of missing values, without relying on imputation for either sensitive or non-sensitive attributes. It is composed of two modules, i.e., an *incomplete data encoding* (IDE) module and a *debiased representation learning* (DRL) module. IDE directly represents incomplete data without imputation. We develop an incomplete tabular embedding strategy and a missingness-aware Transformer block to learn the observed data distribution and missing state information (for addressing CH1). It thus eliminates bias caused by complex data missingness. To enable fairness metric when sensitive attributes are missing, DRL transforms the traditional discrimination risk measurement (based on sensitive attributes) into the difference computation between attention weights (for addressing CH2). It introduces a debiased attention mechanism to reduce the bias in attention weight allocation, thereby minimizing group discrimination. We theoretically prove that, in FATE, the discrimination risk of attention weights is bounded by a constant, significantly reducing group discrimination. In belief, the main contributions of this paper are summarized as follows.

- **Fairness Intervention.** We propose a Transformer-based fairness intervention model FATE that achieves improved accuracy and fairness performance over incomplete data. To the best of our knowledge, it is the first proposal to address the fairness intervention problem over incomplete data that does not rely on the sensitive attribute.
- **Incomplete Data Encoding.** The IDE module of FATE includes an incomplete tabular embedding strategy and a missingness-aware Transformer block to directly learn the observed data distribution and missing state information without imputation. It effectively eliminates bias from complex missingness.

- **Debiased Representation Learning.** The DRL module of FATE introduces a debiased Transformer block to reduce bias in attention weight distribution, thereby minimizing group discrimination and enhancing fairness.
- **Extensive Experiments.** Extensive experiments using various real-world datasets demonstrate that, FATE consistently outperforms the state-of-the-art approaches.

2 Related Work

Traditional fairness intervention methods can be categorized into three paradigms: (1) *pre-processing*, (2) *in-processing*, and (3) *post-processing* approaches. Pre-processing fairness intervention approaches adjust the data distribution to mitigate the association between sensitive attributes and the label, such as FairRF [54], Influence-Fair [27]. In-processing algorithms directly intervene in the learning algorithm with gap regularization [7] or adversarial learning [7] to achieve fairness. Post-processing methods adjust the predictions of the already-trained classifier to ensure fairness, which are model-agnostic [2, 8]. However, the above methods focus on the *full-knowledge* data, and cannot effectively handle common incomplete tabular data.

Algorithmic fairness with missing values can be divided into three categories based on the missing status of sensitive attributes and non-sensitive attributes: i) observed sensitive attributes and partially missing non-sensitive (OS-MN) attributes [24], ii) partially missing sensitive attributes and observed non-sensitive (MS-ON) attributes [45, 53], and iii) completely missing sensitive attributes and observed non-sensitive (CMS-ON) attributes [5, 15, 18]. In particular, FairMIPForset [24] for the OS-MN case employs tree models to directly handle missing data. It incorporates fairness constraints based on sensitive attributes into the optimization objective. For the MS-ON case, they employ contrastive learning [53] or active learning [45] to leverage samples without sensitive attributes, for enhancing fairness. In CMS-ON case, they either utilize proxy attributes for unobserved sensitive [15] or consider the Max-Min fairness as maximizing the minimum utility of any group divided independently of sensitive attributes [5, 18]. Among them, the fairness definition relies on the specific value of sensitive attributes. None of them can directly support our studied problem where both sensitive and non-sensitive attributes are missing.

Imputation algorithms can be classified into three main groups, i.e., statistical, machine learning, and deep learning models. Statistical imputation methods replace the missing value as some statistics [12]. Machine learning imputation approaches train models to estimate the missing values, such as XGBoost imputation [6], MissForest imputation [42]. In contrast, the deep learning imputation methods leverage advanced deep learning models [11] to impute missing values, e.g., MIWAE [31], not-MIWAE [23], GAIN [52], EDIT [32], NOMI [47] and SCIS [50]. As confirmed later, the imputation-then-classification approach introduces additional bias, compromising fairness and accuracy.

3 Problem Definition

In this section, we describe the discrimination risk in group fairness. We then present the problem addressed in our paper and prove that the imputation-then-classification solution is less effective.

Given an incomplete tabular dataset denoted as $\mathcal{D} = \{\mathbf{X}, \mathbf{S}, \mathbf{Y}\} = \{\mathbf{x}_i, \mathbf{s}_i, \mathbf{y}_i\}_{i=1}^n$, where $\mathbf{X}$ includes all non-sensitive attributes, $\mathbf{S}$ represents the sensitive attributes (e.g., sex and race), and $\mathbf{Y}$ denotes the label set. Each data sample $\mathbf{x}_i = (x_{i1}, \dots, x_{ij}, \dots, x_{id})$ with x_{ij} takes values from the feature space $\mathcal{X}_j$, and d represents the total number of features. To represent the missing information in $\mathbf{X}$, a mask matrix $\mathbf{M}$ of $\mathbf{X}$ is defined as $\mathbf{M} = (\mathbf{m}_1, \dots, \mathbf{m}_n)^\top$. Each mask vector $\mathbf{m}_i = (m_{i1}, \dots, m_{id})$ corresponds to a sample $\mathbf{x}_i$. m_{ij} takes value from $\{0, 1\}$, $m_{ij} = 0$ (resp. 1) iff the j -th dimension is observed (resp. missing). As a new effort in this field, we are focusing on the binary scenario, where both target label and sensitive attribute are binary variables, i.e., $y_i \in \{0, 1\}$ and $s_i \in \{0, 1\}$.

Algorithmic fairness means that, the algorithm deals with all groups and individuals in a non-biased manner, without favoring or discriminating any particular group or individual based on sensitive attributes. In this paper, we focus on *group fairness*. Group fairness *requires* that a decision model $\mathcal{H}$ performs consistently across different groups defined by *sensitive attributes* (e.g., gender, age, or race). For the decision model $\mathcal{H}$ dealing with a single binary sensitive attribute, we define the *discrimination risk* as

$$\text{Disc}(\mathcal{H}) = |\Pr_{s=0}(\mathcal{H}) - \Pr_{s=1}(\mathcal{H})|, \quad (1)$$

where $\Pr_{s=0}(\mathcal{H})$ (resp. $\Pr_{s=1}(\mathcal{H})$) means the prediction performance in demographic group 0 (resp. demographic group 1).

The discrimination risk measures the performance difference among different demographic groups. $\text{Disc}(\mathcal{H}) = 0$ means that, the model is fair. $\text{Disc}(\mathcal{H}) > 0$ indicates that, the model is discriminatory. We can utilize various metrics to compute $\Pr_{s=0}(\mathcal{H})$ (resp. $\Pr_{s=1}(\mathcal{H})$) and assess fairness, such as equality of opportunity or equalized odds [17].

The goal of our studied problem in this paper is to train a classifier $\mathcal{H}$ that utilizes the incomplete tabular data $\mathbf{X}$ with missing (both sensitive and non-sensitive) attributes to predict the label $\mathbf{Y} \in \{0, 1\}$. We formulate the studied problem in this paper in Definition 1.

DEFINITION 1. (Fairness-aware prediction with incomplete data). Given a dataset $\mathcal{D}$ that contains incomplete non-sensitive attributes $\mathbf{X}$ with its mask matrix $\mathbf{M}$ and fully missing sensitive attributes $\mathbf{S}$, the goal of fairness-aware prediction over $\mathcal{D}$ is to train a classification model $\mathcal{H} : \mathbb{R}^d \mapsto \mathcal{Y}$, where $\mathcal{Y}$ denotes the discrete set of the label $\mathbf{Y}$, such that it maintains high accuracy with bounded discriminatory risk, i.e., $\max_{\mathcal{H}} \mathbb{E}[\mathbb{I}(\mathcal{H}(\mathbf{X}) = \mathbf{Y})]$ subject to $\text{Disc}(\mathcal{H}) \leq \epsilon$, where $\epsilon \geq 0$ is a tolerance threshold, $\mathbb{I}$ is the indicator function, returning 1 if $\mathcal{H}$ correctly predicts $\mathcal{H}(\mathbf{X}) = \mathbf{Y}$, and 0 otherwise.

The optimal solution $F_\epsilon(\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}})$ of this problem only depends on the data distribution $\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}}$ and ϵ . Based on probability theory, we observed that, following model optimization, the analysis method that directly utilizes incomplete data is more effective than the imputation-then-classification solution, as highlighted in Observation 1. Our theoretical analysis leads to the conclusion that in complex missing data scenarios, the imputation-then-classification approach can lose accuracy and compromise fairness.

OBSERVATION 1. Suppose that an incomplete dataset $\mathcal{D}$ comprises $\mathbf{X}$ containing two features, a label set $\mathbf{Y}$, and a single set of sensitive attribute $\mathbf{S}$. Let α denote the proportion of positive samples in the dataset. For any ϵ and $\alpha \in (0, \frac{1}{3})$, there exists a data distribution

$\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}}$ such that the optimal solution $\mathcal{H}_\epsilon(\Pr_{\hat{\mathbf{X}}, \mathbf{S}, \mathbf{Y}})$ is less than or equal to the optimal solution $\mathcal{H}_\epsilon(\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}})$, i.e.,

$$\sup_{f_{\text{imp}}} \mathcal{H}_\epsilon(\Pr_{\hat{\mathbf{X}}, \mathbf{S}, \mathbf{Y}}) \leq \mathcal{H}_\epsilon(\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}}) - \frac{1 - \alpha}{2}. \quad (2)$$

In particular, f_{imp} is the mapping function of the imputation model. The imputed data is represented as $\hat{\mathbf{X}} = f_{\text{imp}}(\mathbf{X})$.

PROOF. We prove this observation through construction. Let us consider a dataset $\mathbf{X}$ with two features, $\mathbf{x}_1$ and $\mathbf{x}_2$, which can take values from the set $\{0, 1\}$. Assume that the feature $\mathbf{x}_1$ has missing values, which is missing not at random. We construct the probability distribution $\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}}$ as follows.

$$\begin{aligned} \Pr(\mathbf{Y} = 0, \mathbf{X} = (0, 0) | \mathbf{S} = s) &= \Pr(\mathbf{Y} = 0, \mathbf{X} = (0, 1) | \mathbf{S} = s) = \frac{1 - \alpha_s}{4}, \\ \Pr(\mathbf{Y} = 0, \mathbf{X} = (1, 0) | \mathbf{S} = s) &= \Pr(\mathbf{Y} = 0, \mathbf{X} = (1, 1) | \mathbf{S} = s) = \frac{1 - \alpha_s}{4}, \\ \Pr(\mathbf{Y} = 1, \mathbf{X} = (0, 0) | \mathbf{S} = s) &= \Pr(\mathbf{Y} = 1, \mathbf{X} = (0, 1) | \mathbf{S} = s) = 0, \\ \Pr(\mathbf{Y} = 1, \mathbf{X} = (1, 0) | \mathbf{S} = s) &= \Pr(\mathbf{Y} = 1, \mathbf{X} = (1, 1) | \mathbf{S} = s) = 0, \\ \Pr(\mathbf{Y} = 0, \mathbf{X} = (\text{NA}, 0) | \mathbf{S} = s) &= \Pr(\mathbf{Y} = 0, \mathbf{X} = (\text{NA}, 1) | \mathbf{S} = s) = 0, \\ \Pr(\mathbf{Y} = 1, \mathbf{X} = (\text{NA}, 0) | \mathbf{S} = s) &= \Pr(\mathbf{Y} = 1, \mathbf{X} = (\text{NA}, 1) | \mathbf{S} = s) = \frac{\alpha_s}{2}. \end{aligned}$$

We represent a probabilistic classifier $\mathcal{H}$ such that $\mathcal{H}(\mathbf{X} = (0, 0)) = \mathcal{H}(\mathbf{X} = (0, 1)) = \mathcal{H}(\mathbf{X} = (1, 0)) = \mathcal{H}(\mathbf{X} = (1, 1)) = 0$, $\mathcal{H}(\mathbf{X} = (\text{NA}, 0)) = \mathcal{H}(\mathbf{X} = (\text{NA}, 1)) = 1$. In this situation, $\mathbb{E}[\mathbb{I}(\mathcal{H}(\mathbf{X}) = \mathbf{Y})] = 1$, and $\text{Disc}(\mathcal{H}) = 0$. For any ϵ , $\mathcal{H}_\epsilon(\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}}) = 1$. By symmetry, we assume $f_{\text{imp}}(\text{NA}) = 1$ without loss of generality. We represent a probabilistic classifier $\mathcal{H}$ as

$$\begin{aligned} \Pr(\hat{\mathbf{Y}} = 0 | \hat{\mathbf{X}} = (0, 0)) &= \Pr(\hat{\mathbf{Y}} = 0 | \hat{\mathbf{X}} = (0, 1)) = \frac{p_0}{2}, \\ \Pr(\hat{\mathbf{Y}} = 0 | \hat{\mathbf{X}} = (1, 0)) &= \Pr(\hat{\mathbf{Y}} = 0 | \hat{\mathbf{X}} = (1, 1)) = \frac{p_1}{2}, \\ \Pr(\hat{\mathbf{Y}} = 1 | \hat{\mathbf{X}} = (0, 0)) &= \Pr(\hat{\mathbf{Y}} = 1 | \hat{\mathbf{X}} = (0, 1)) = \frac{1 - p_0}{2}, \\ \Pr(\hat{\mathbf{Y}} = 1 | \hat{\mathbf{X}} = (1, 0)) &= \Pr(\hat{\mathbf{Y}} = 1 | \hat{\mathbf{X}} = (1, 1)) = \frac{1 - p_1}{2}. \end{aligned}$$

Consequently, we have

$$\begin{aligned} \Pr(\hat{\mathbf{Y}} = 0 | \mathbf{Y} = 1, \mathbf{S} = s) &= \frac{p_1}{2}, \Pr(\hat{\mathbf{Y}} = 1 | \mathbf{Y} = 0, \mathbf{S} = s) = \frac{2 - p_0 - p_1}{4}, \\ \Pr(\hat{\mathbf{Y}} = 0 | \mathbf{Y} = 0, \mathbf{S} = s) &= \frac{p_0 + p_1}{4}, \Pr(\hat{\mathbf{Y}} = 1 | \mathbf{Y} = 1, \mathbf{S} = s) = \frac{1 - p_1}{2}. \end{aligned}$$

This ensures that $\text{Disc}(\mathcal{H}) = 0$. Moreover, we can obtain $\mathbb{E}[\mathbb{I}(\hat{\mathbf{Y}} = \mathbf{Y})] = \frac{1 - \alpha}{4} p_0 + \frac{1 - 3\alpha}{4} p_1 + \frac{\alpha}{2}$.

Given that $\alpha \in (0, 1/3)$, it follows that $1 - \alpha > 0$ and $1 - 3\alpha > 0$. Consequently, the objective function attains its maximum value when $p_0 = p_1 = 1$, resulting in $(1 - \alpha)/2$. In other words, $H_\epsilon(\Pr_{\hat{\mathbf{X}}, \mathbf{S}, \mathbf{Y}}) = (1 - \alpha)/2$. Since $F_\epsilon(\Pr_{\mathbf{X}, \mathbf{S}, \mathbf{Y}}) = 1$, (2) holds. Therefore, the imputation-then-classify solution is less effective, compared to the analysis method using incomplete data without imputation. $\square$

Observation 1 reveals that, the imputation-then-classification approach will introduce additional bias, harming the accuracy of the algorithm, compared to the analysis method directly using incomplete data without imputation. Moreover, the decline in accuracy becomes increasingly evident as the dataset becomes imbalanced.

4 Solution Overview

Figure 1 depicts the framework of our proposed Transformer-based fairness-aware prediction model FATE over incomplete data.

Inspired by recent advances in Transformer-based models for tabular data (e.g., Tabtransformer[21], SAINT[41]), which leverage self-attention to capture complex dependencies among features, we propose a novel Transformer-based fairness-aware prediction

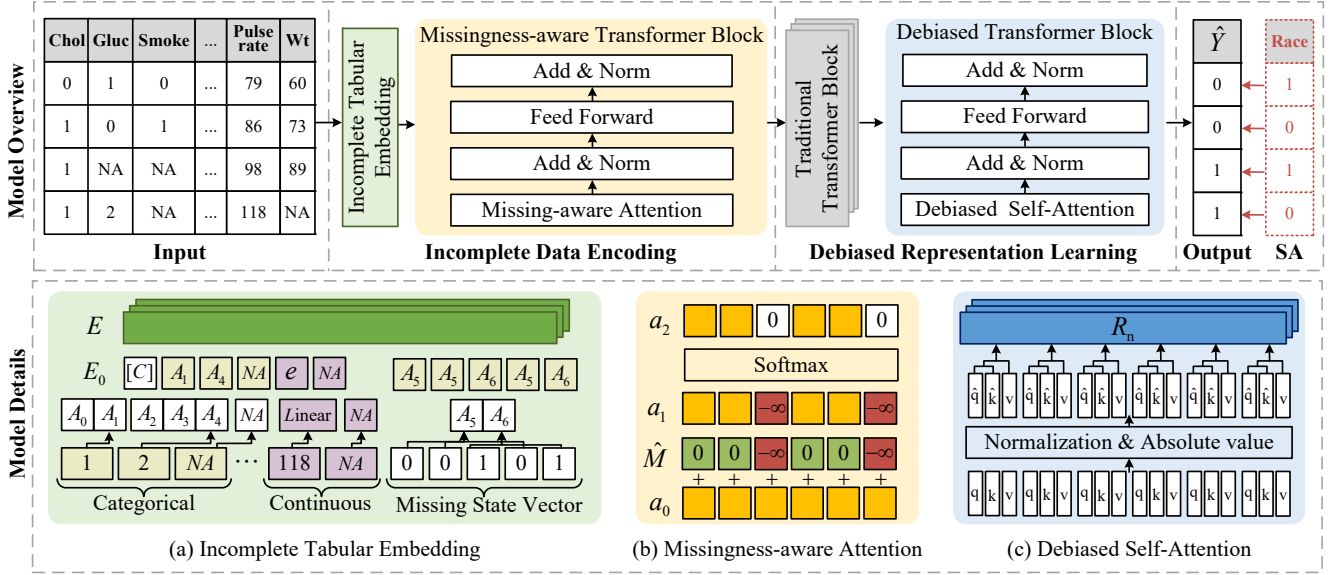


Figure 1: The architecture of FATE

model FATE. FATE consists of an *incomplete data encoding* (IDE) module and a *debiased representation learning* (DRL) module. IDE directly represents incomplete data without using sensitive attributes, and thus eliminates bias caused by complex data missingness. DRL transforms the traditional discrimination risk measurement (based on sensitive attributes) into an efficient difference computation between attention weights, enabling optimization of algorithmic fairness without sacrificing accuracy.

Taking the incomplete attribute matrix $X \in \mathbb{R}^{N \times d}$ (that contains numerical and categorical non-sensitive attributes) as input, FATE computes the fair-aware prediction $\hat{Y}$ using the optimized classification model $\mathcal{H}^*$. Specifically, the IDE module of FATE proposes an *incomplete tabular embedding* strategy that concatenates the non-sensitive attribute matrix X with its corresponding mask matrix M . Then, it employs a separate single full-connected layer with a ReLU nonlinearity to embed each continuous attribute in X independently. Finally, the IDE module designs a missingness-aware Transformer block that utilizes a missingness-aware attention mechanism to learn both the observed data distribution and missing state distribution from incomplete tabular data for more effective representation.

The DRL module of FATE first generates an informative incomplete data representation using the traditional Transformer blocks. It then employs a novel debiased Transformer block with a carefully debiased self-attention mechanism, incorporating normalization and absolute value operations, to reduce group discrimination and enhance fairness. In this way, the DRL module generates an unbiased incomplete data representation that preserves predictive power while ensuring fair classification.

5 FATE Details

In this section, we elaborate on the two modules of incomplete data encoding (IDE) and debiased representation learning (DRL) in the FATE model, respectively.

5.1 Incomplete Data Encoding

The incomplete data encoding (IDE) module of FATE aims to capture both the observed data distribution and missing state information in incomplete tabular data. The IDE module adopts an *incomplete tabular embedding* strategy and a *missingness-aware Transformer* block to achieve this dual objective through innovative architectural design. Specifically, the IDE module introduces the incomplete tabular embedding strategy to concatenate the input data matrix X with the mask matrix M . For categorical features, we employ an embedding table to project each discrete value into a r -dimensional latent space. For continuous features, we apply a fully-connected layer with ReLU activation to transform each numerical attribute into the r -dimensional embedding space. The missingness-aware Transformer blocks utilize the missing-aware attention mechanism to directly learn the observed data distribution and the missing state distribution without imputation.

Incomplete tabular embedding. As demonstrated in Observation 1, the missing state is highly informative with respect to the predictive label. To retain the rich information encoded in the missingness of the incomplete tabular data, we include the missing state matrix M as a part of the input. Specifically, for the non-sensitive attributes X in the incomplete dataset and its corresponding missing state matrix M , we concatenate them along the features as input. This concatenation enables the model to directly grasp the missing state information within the incomplete tabular data [13], preserving its predictive utility. For categorical features, we design a look-up table to project each discrete value into a r -dimensional space. Moreover, inspired by [41], we use a fully-connected layer with a ReLU nonlinearity to map each continuous feature into a r -dimensional space, so that the continuous features can be fed into the Transformer block. Instead of conventional imputation of missing values, we introduce a r -dimensional padding vector to represent missing values. They are then concatenated as the embedding matrix E .

Missingness-aware Transformer block. It is worthwhile to point out that, traditional Transformer models require additional imputation algorithms to handle missing data. Even when using padding vectors to replace the embeddings for missing values, it can still introduce biases in calculating attention scores. In light of this, we incorporate a missingness-aware Transformer (MT) block with the missing-aware attention mechanism. This innovative block enables the model to simultaneously learn the distributions of both observed data and the missing state of incomplete data. It facilitates effective representation generation.

Specifically, on top of the embedding matrix E , the MT block first computes the query Q , key K , and value V matrices, respectively, by utilizing a linear neural network. This network maps the embedding into a reduced dimension space d_k . Based on the mask matrix M , we generate the masking matrix $\hat{M}$, where each element $\hat{m}_{ij} \in \hat{M}$ takes a value from $\{0, -\infty\}$. Specifically, $\hat{m}_{ij} = 0$ (resp. $-\infty$) iff the $m_{ij} = 0$ (resp. 1). Then, the attention matrix A can be calculated.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} + \hat{M} \right) V. \quad (3)$$

The missingness-aware attention mechanism uses the masking matrix $\hat{M}$ that adds $-\infty$ to the attention scores corresponding to the missing values. The attention scores taking values of $-\infty$ will be ignored after the *softmax* operation, since they will become zero. This mechanism enables the model to learn directly from incomplete data without imputation, effectively eliminating the influence of missing values on data representation.

In the following, the attention score matrix A is transformed by a linear layer to match the dimensions of the input. Next, the latent representations R are average-pooled, i.e., $R = \text{Avg-pool}(W \odot A)$, where Avg-pool is the average-pooling operation, W is the parameter matrix to be learned. Finally, the IDE module generates its effective representation R_{id} for the incomplete tabular dataset.

5.2 Debaised Representation Learning

DRL aims at optimizing the algorithmic fairness with the absence of sensitive attributes. The findings [35, 43] show that, addressing fairness concerns within the attention mechanism is effective and outperforms other in-processing methods. The disparities in average query attention across different sensitive groups [29, 35] reveal potential biases in the attention mechanism.

According to Eq. 3, the attention mechanism can be divided into two stages: i) allocate attention weights W via the inner product of Q and K , and ii) perform a weighted summation over V . Discrimination in transformer-based models can stem from the misallocation of attention weights.

Inspired by fairness representation [16, 28], in the *debaised representation learning* (DRL) module, we present a debaised self-attention mechanism to reduce fairness bias in tabular data representation. Specifically, we denote w_{ij} as the attention weight of the j -th feature on sample x_i . Given the fairness consideration, our optimization problem can be written as

$$\min \sum_{i=1}^n \sum_{j=1}^d \mathbb{E}[w_{ij}|s=0] - \mathbb{E}[w_{ij}|s=1]. \quad (4)$$

The goal of the DRL module is to minimize the disparity in attention weights W among sensitive groups for each feature dimension. Directly addressing this issue through Eq. 4 is challenging, due to the unavailability of sensitive attribute information. Theorem 1 claims that, a series of operations can effectively reduce disparities in attention allocation among different sensitive groups.

ASSUMPTION 1. Given a dataset X that only has a single discrete feature, $x_i \in \{0, 1\}$, and the sensitive attribute S is a binary attribute, $s_i \in \{0, 1\}$. The query q and key k follows a Gaussian distribution, denotes as $P(q | s = s_i, x = x_i) = \mathcal{N}(\mu_{q_{s_i, x_i}}, \sigma_{q_{s_i, x_i}})$ and $P(k | s = s_i, x = x_i) = \mathcal{N}(\mu_{k_{s_i, x_i}}, \sigma_{k_{s_i, x_i}})$. The distribution of the feature and sensitive attribute is $P(x = 0, s = 0) = \lambda_0$, $P(x = 0, s = 1) = \lambda_1$, $P(x = 1, s = 0) = \lambda_2$, and $P(x = 1, s = 1) = \lambda_3$. The distribution of the query q and the key k in the transformer can be defined as

$$\begin{aligned} q &\sim \lambda_0 \mathcal{N}(\mu_{q_{0,0}}, \sigma_{q_{0,0}}) + \lambda_1 \mathcal{N}(\mu_{q_{0,1}}, \sigma_{q_{0,1}}) \\ &\quad + \lambda_2 \mathcal{N}(\mu_{q_{1,0}}, \sigma_{q_{1,0}}) + \lambda_3 \mathcal{N}(\mu_{q_{1,1}}, \sigma_{q_{1,1}}), \\ k &\sim \lambda_0 \mathcal{N}(\mu_{k_{0,0}}, \sigma_{k_{0,0}}) + \lambda_1 \mathcal{N}(\mu_{k_{0,1}}, \sigma_{k_{0,1}}) \\ &\quad + \lambda_2 \mathcal{N}(\mu_{k_{1,0}}, \sigma_{k_{1,0}}) + \lambda_3 \mathcal{N}(\mu_{k_{1,1}}, \sigma_{k_{1,1}}). \end{aligned} \quad (5)$$

THEOREM 1. For a feature $x = [x_1, \dots, x_i, \dots, x_n]$,

$$n(x) = \left[\frac{x_1 - \mathbb{E}[x]}{\sqrt{\text{Var}[x]}}, \dots, \frac{x_n - \mathbb{E}[x]}{\sqrt{\text{Var}[x]}} \right]. \quad (6)$$

and $\text{abs}(x) = [|x_1|, \dots, |x_n|]$. Given query $q \in \mathbb{R}^{d_k}$ and key $k \in \mathbb{R}^{d_k}$, the debaised query and key is denoted as $q^{de} = m \circ n(q) \in \mathbb{R}^{d_k}$ and $k^{de} = m \circ n(k) \in \mathbb{R}^{d_k}$. The disparity in debaised attention weights δ^{de} is upper bounded by

$$\begin{aligned} |\delta^{de}| &\leq \sqrt{d_k} \left[\left(\sqrt{\frac{2}{\pi\lambda_0}} + \frac{\lambda_1}{\lambda_0} \right)^2 + \left(\sqrt{\frac{2}{\pi\lambda_1}} + \frac{\lambda_0}{\lambda_1} \right)^2 \right] \\ &\quad + \sqrt{d_k} \left[\left(\sqrt{\frac{2}{\pi\lambda_2}} + \frac{\lambda_3}{\lambda_2} \right)^2 + \left(\sqrt{\frac{2}{\pi\lambda_3}} + \frac{\lambda_2}{\lambda_3} \right)^2 \right]. \end{aligned} \quad (7)$$

PROOF. Based on the Assumption 1, the expectation of q is

$$\begin{aligned} \mu_{x=0, s=j} &= \mathbb{E}[q|x=0, s=0] + \mathbb{E}[q|x=0, s=1] = \lambda_0 \mu_0 + \lambda_1 \mu_1, \\ \mu_{x=1, s=j} &= \mathbb{E}[q|x=1, s=0] + \mathbb{E}[q|x=1, s=1] = \lambda_2 \mu_2 + \lambda_3 \mu_3, \end{aligned}$$

and the variance can be written as

$$\begin{aligned} \sigma_{x=0, s=j}^2 &= (\lambda_0 \sigma_{0,0}^2 + \lambda_1 \sigma_{0,1}^2) + \lambda_0 \lambda_1 \Delta^2, \quad \text{let } \Delta_0 = \mu_{0,0} - \mu_{0,1} > 0. \\ \sigma_{x=1, s=j}^2 &= (\lambda_2 \sigma_{1,0}^2 + \lambda_3 \sigma_{1,1}^2) + \lambda_2 \lambda_3 \Delta^2, \quad \text{let } \Delta_1 = \mu_{1,0} - \mu_{1,1} > 0. \end{aligned}$$

After normalization with $q^{norm} = \frac{q - \mathbb{E}[q]}{\sqrt{\text{Var}[q]}}$, the conditioned variable $q^{norm}|x = i, s = j$ are still Gaussian, and after taking the absolute values, the conditioned expectations can be written as

$$\begin{aligned} \mathbb{E}[q^{de} | x = i, s = j] &= - \int_{-\infty}^0 t p(t) dt + \int_0^{\infty} t p(t) dt \\ &= \sigma_{i,j}^{norm} \sqrt{\frac{2}{\pi}} \exp \left(- \frac{(\mu_{i,j}^{norm})^2}{2(\sigma_{i,j}^{norm})^2} \right) + \mu_{i,j}^{norm} \text{erf} \left(\frac{\mu_{i,j}^{norm}}{\sqrt{2}\sigma_{i,j}^{norm}} \right), \end{aligned}$$

where $p(t) = \mathcal{N}(\mu_{i,j}^{norm}, (\sigma_{i,j}^{norm})^2)$, $i = \{0, 1\}$, $j = \{0, 1\}$. Then the expectation can be bounded by

$$\mathbb{E}[q^{de} | x = 0, s = j] \leq \frac{1}{\sigma} \left[\sigma_{0,j} \sqrt{\frac{2}{\pi}} + \lambda_{1-j} \Delta_0 \right] = \sqrt{\frac{2}{\pi\lambda_j}} + \sqrt{\frac{\lambda_{1-j}}{\lambda_j}}.$$

Algorithm 1: The Procedure of FATE

Input: an incomplete data matrix $\mathbf{X}$ with its mask matrix $\mathbf{M}$, a transformer-based model $\mathcal{H}$, the label $\mathbf{Y}$, training times t ;

Output: the trained model $\mathcal{H}_*$ and predict label $\hat{\mathbf{Y}}$

```

1: for each feature  $\mathbf{x}_j$  in  $\mathbf{X}$  do
2:   /* Incomplete Tabular Embedding */
3:   if  $\mathbf{x}_j$  is category feature then
4:     lookup the embedding table to get embedding  $\mathbf{e}_j^{\text{cat}}$ 
5:   else if  $\mathbf{x}_j$  is continuous feature then
6:     use a fully-connected layer to get embedding  $\mathbf{e}_j^{\text{num}}$ 
7: for epoch = 1 to  $t$  do
8:   if epoch == 1 then
9:     /* Missingness-aware Transformer Block */
10:    generate the attention matrix  $\hat{\mathbf{M}}$  based on  $\mathbf{M}$ 
11:    calculate the attention score
12:     $\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \hat{\mathbf{M}}\right) \mathbf{V}$ 
13:   else if epoch ==  $t$  then
14:     /* Debiased Representation Learning */
15:    calculate the debiased query  $\mathbf{Q}^{\text{de}}$  and key  $\mathbf{K}^{\text{de}}$ 
16:    calculate the debiased attention score
17:     $\mathbf{A}^{\text{de}} = \text{softmax}\left(\frac{\mathbf{Q}^{\text{de}}\mathbf{K}^{\text{de}T}}{\sqrt{d_k}}\right) \mathbf{V}$ 
18:   else
19:     /* Traditional Transformer Block */
20:    calculate the attention score  $\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}$ 
21:    calculate the supervised loss  $\mathcal{L}$ 
22:    update the model  $\mathcal{H}$ 's parameters using  $\mathcal{L}$ 
23: return  $\mathcal{H}_*$  and  $\hat{\mathbf{Y}}$ 

```

Similarly, we can obtain result for $\mathbf{k}$. Now the original statement can be re-written as

$$|\mu_{q_0^{\text{de}}} \mu_{k_0^{\text{de}}} - \mu_{q_1^{\text{de}}} \mu_{k_1^{\text{de}}}| \leq \left(\sqrt{\frac{2}{\pi\lambda_0}} + \sqrt{\frac{\lambda_1}{\lambda_0}}\right)^2 + \left(\sqrt{\frac{2}{\pi\lambda_1}} + \sqrt{\frac{\lambda_0}{\lambda_1}}\right)^2 + \left(\sqrt{\frac{2}{\pi\lambda_2}} + \sqrt{\frac{\lambda_3}{\lambda_2}}\right)^2 + \left(\sqrt{\frac{2}{\pi\lambda_3}} + \sqrt{\frac{\lambda_2}{\lambda_3}}\right)^2.$$

□

Theorem 1 indicates that, the upper bound of the expected difference in attention weights could be bounded by a constant. The constant depends on feature distribution and embedding dimension.

It is worth noting that, without the normalization and absolute value operation, the expected difference in attention weights can be computed as

$$|\delta| = \frac{1}{\sqrt{d_k}} \sum_{x_i \in \{0,1\}} \sum_{k_i}^d |(\mu_{q_{0,x_i}} \mu_{k_{0,x_i}} - \mu_{q_{1,x_i}} \mu_{k_{1,x_i}})|. \quad (8)$$

Through Eq. 8, we can observe that, the expected difference in attention weights, δ , is divergent toward infinity and can vary

significantly with changes in the distributions of $\mathbf{q}$ and $\mathbf{k}$. A common approach to improve the fairness of attention weight is to constrain δ within a constant β range and include it in the optimization objective. However, the sensitive attribute S is unknown during the training phase, so we cannot directly calculate δ . Thus, in DRL, we employ the normalization and absolute value operation in the difference computation.

Objective. The objective of FATE is to minimize the cross-entropy loss for downstream classification tasks. Since sensitive attributes are completely missing, we do not incorporate a fairness constraint into the loss function. Instead, we found that the debiasing self-attention mechanism in the DRL module is sufficient to reduce group discrimination. Thus, the objective function of FATE is

$$\mathcal{L} = \sum_{i=1}^n \sum_{j=1}^C y_{ij} \log(\hat{y}_{ij}), \quad (9)$$

where n is the sample size and C is the number of classes in $\mathbf{Y}$.

The pseudo-code for FATE is presented in Algorithm 1. It takes an incomplete data matrix $\mathbf{X}$ with its mask matrix $\mathbf{M}$ as input, and outputs the optimized fair prediction model $\mathcal{H}_*$. Specifically, for each feature x_j , we first acquire its embedding representation according to the feature type (Lines 1-5). During model training, the training epochs are divided into three distinct phases. In the first epoch, the missing aware Transformer block generates the attention matrix $\hat{\mathbf{M}}$ based on $\mathbf{M}$ and we compute the attention score to learn data representation from incomplete data (Lines 7-9). In the last epoch t , we use the debiased attention score to conduct debiased representation learning (Lines 10-12). For all other epochs, standard Transformer operations are carried out (Line 12). After each epoch, we compute the supervised loss and update the parameters (Lines 14-15). Finally, the algorithm outputs the optimized model $\mathcal{H}_*$ and corresponding predictions $\hat{\mathbf{Y}}$.

6 Experiments

In this section, we evaluate the performance of our proposed algorithms in different aspects. All algorithms were implemented with Python and conducted on an Intel Core 2.80GHz Server with 512GB RAM, running Ubuntu 18.04 system.

Datasets. We conduct experiments on three public tabular datasets. The *Adult* dataset [3] contains 48,842 samples with 14 features, surveying whether an individual's income exceeds \$50K per year. Sex is the sensitive attribute in *Adult*. The high school longitudinal study (*HSLs*) dataset [22] is a survey of over 23,000 high school students, followed from 9-th through 12-th grade, to predict the student's 9th-grade math test scores. We use a subset of the dataset containing 21,444 samples with 9 features, following the methodology described in [13]. The sensitive attribute in this dataset is race. The American Community Survey public use Microdata sample (ACS) dataset [10] is a newly proposed dataset from the American Community Survey. We consider the income prediction task over ACS, which contains 195,665 samples with 9 features. The sensitive attribute is gender. For all datasets, we randomly choose 10% for testing, 10% for validation, and the rest for training.

Missing mechanism simulation. We inspect three missing mechanisms MCAR, MAR, and MNAR for evaluation. The missing values are randomly injected to $\mathbf{X}$ to simulate the default missing

Table 1: The prediction performance under MAR

Method	Imputation	Adult			HSLs			ACS		
		ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓
DRO	GAIN	0.6650	0.1102	0.3204	0.5477	0.0380	0.0889	0.5967	0.0349	0.0569
	Mean	0.7368	0.1744	0.3892	0.5855	0.0253	<u>0.0434</u>	0.6250	0.0355	0.0679
	MICE	0.7138	0.1182	0.2329	0.5872	0.0351	0.0560	0.6089	0.0301	0.0556
	MIWAE	0.7145	0.1127	0.2657	0.5891	0.0270	0.0451	0.6114	0.0177	0.0497
	NOMI	0.7298	0.1067	0.2732	0.5692	0.0288	0.0643	0.6371	0.0153	0.0658
	notMIWAE	0.7169	0.1002	0.2402	0.5758	0.0320	0.0758	0.6127	0.0162	0.0502
	Zero	0.6964	0.1083	0.2598	0.5680	0.0320	0.0602	0.6192	0.0356	0.0523
FairRF	GAIN	0.6747	0.1296	0.1585	0.5534	0.0297	0.0936	0.5995	0.0206	0.0488
	Mean	0.7025	0.1416	<u>0.1557</u>	0.5854	0.0560	0.1377	0.6185	0.0301	0.0580
	MICE	0.7280	0.1424	0.1713	0.5799	0.0706	0.1463	0.6123	0.0313	0.0488
	MIWAE	0.7064	0.1752	0.1954	0.5790	0.0478	0.1195	0.6081	0.0206	0.0458
	NOMI	0.7384	0.2380	0.2611	0.5795	0.0753	0.1561	0.6124	0.0186	0.0507
	notMIWAE	0.7005	0.1609	0.1843	0.5763	0.0334	0.0927	0.6116	0.0250	0.0501
	Zero	0.7068	0.1673	0.1742	0.5782	<u>0.0262</u>	0.0384	0.5878	0.0194	<u>0.0421</u>
SURE	GAIN	0.7355	0.1332	0.3772	0.5916	0.0610	0.1033	0.6201	<u>0.0106</u>	0.0481
	Mean	0.7554	0.1781	0.4277	0.6108	0.0715	0.1372	0.6307	0.0167	0.0544
	MICE	0.7559	0.1510	0.3447	0.6151	0.0673	0.1416	0.6264	0.0243	0.0615
	MIWAE	<u>0.7730</u>	<u>0.0789</u>	0.1599	0.6028	0.0554	0.1079	0.6218	0.0183	0.0434
	NOMI	0.7580	0.1169	0.3450	0.6056	0.0723	0.1418	<u>0.6416</u>	0.0128	0.0655
	notMIWAE	0.7522	0.0963	0.3199	0.6028	0.0672	0.1441	0.6209	0.0208	0.0480
	Zero	0.7440	0.0668	0.2386	<u>0.6198</u>	0.0417	0.0762	0.6327	0.0143	0.0428
FATE	–	0.7892	0.0944	0.1534	0.6522	0.0469	0.0672	0.7534	0.0102	0.0251

Table 2: The prediction performance under MCAR

Method – Imputation	Adult			HSLs			ACS		
	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓
DRO – NOMI	0.7206	<u>0.1354</u>	0.2689	0.5847	<u>0.0509</u>	<u>0.0958</u>	0.6164	0.0216	0.0538
FairRF – NOMI	0.7125	0.2089	<u>0.2393</u>	0.5867	0.0633	0.1532	0.6262	<u>0.0085</u>	<u>0.0503</u>
SURE – NOMI	<u>0.7517</u>	0.1958	0.4139	<u>0.6030</u>	0.0822	0.1367	<u>0.6286</u>	0.0179	0.0593
FATE	0.7850	0.0966	0.1536	0.6510	0.0349	0.0622	0.7522	0.0066	0.0224

mechanism MCAR. We follow the methodology in [9] to simulate MAR. It first selects a numerical attribute f_v . For a sample $\mathbf{x}_i$, the value x_{ij} in a feature f_j is missing at the probability of $P_c(x_{ij})$, which is calculated according to the value (denoted by x_{iv}) of $\mathbf{x}_i$ on f_v . Namely, $P_c(x_{ij}) = \Phi(x_{iv}) / \sum_{i=1}^n \Phi(x_{iv})$, where $\Phi(x_{iv})$ denotes the ranking of x_{iv} in f_v . The smaller the value in f_v , the higher the ranking. Similar as [44], to simulate MNAR, the value x_{im} of a sample $\mathbf{x}_i$ is missing at the probability of $P_m(x_{im})$, i.e., $P_m(x_{im}) = \Phi(x_{im}) / \sum_{i=1}^n \Phi(x_{im})$.

Baselines. We compare our approach with three state-of-the-art methods. (i) DRO [18] uses distributional robust optimization to achieve fairness without relying on sensitive attributes. It minimizes the worst-case risk overall distributions close to the empirical distribution. (ii) FairRF [57] utilizes the related features to learn fair classifiers and adjusts the importance weights of each related feature for regularization. (iii) SURE [5] is an iterative algorithm that identifies high-risk zones in feature space where the risk of unfair classification is statistically significant. It achieves a fair classifier by adjusting the loss function’s weights for points from these zones. Since all of these models cannot be trained with incomplete tabular data, we employ *seven* state-of-the-art data imputation methods, i.e., Mean, Zero, MICE [38], MIWAE [31], not-MIWAE [23], GAIN

[52], NOMI [47], to impute missing feature values. We do not compare with the algorithm for the case of OS-MN attributes [24], as it requires the sensitive attribute during the optimization process.

Evaluation metrics. We report *Accuracy*(ACC) to assess classification performance. To evaluate fairness, we adopt two widely used group fairness metrics from prior work [13, 57]: *Equality of Opportunity* (EOPP) and *Equalized Odds* (EODD). For binary classification (i.e., $Y = 0$ or $Y = 1$) on two sensitive groups (i.e., $S = 0$ or $S = 1$), the fairness disparities Δ_{EOPP} and Δ_{EODD} are defined as:

$$\Delta_{EOPP} = \left| \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1) \right|,$$

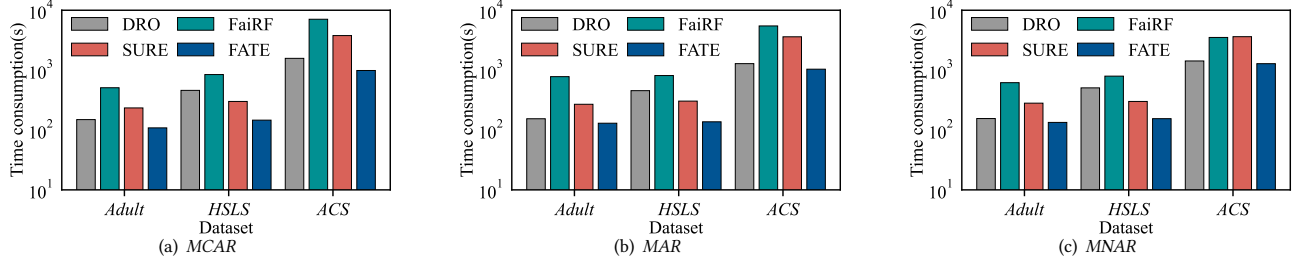
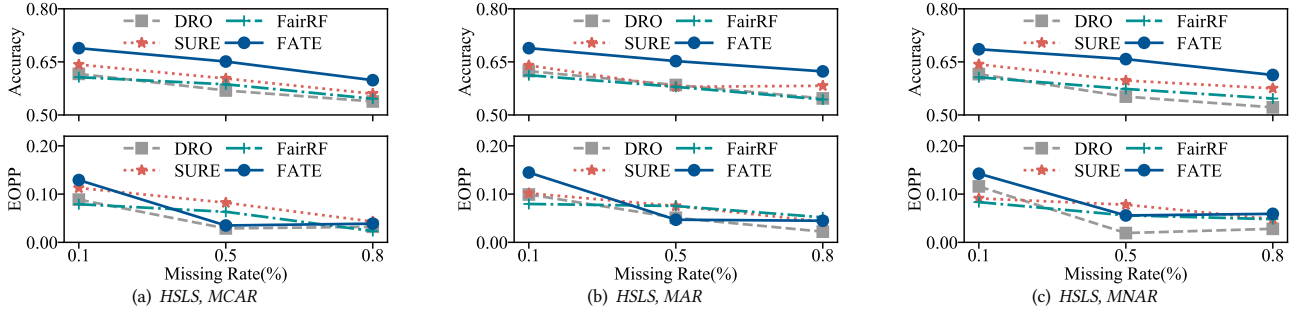
$$\Delta_{EODD} = \left| \mathbb{P}(\hat{Y} = 1 \mid Y = y, S = 0) - \mathbb{P}(\hat{Y} = 1 \mid Y = y, S = 1) \right|.$$

A model is considered fairer when Δ_{EOPP} and Δ_{EODD} are smaller.

Implementation details. Through comprehensive grid search, we optimized the hyperparameters. Specifically, we set the learning rate as 0.3 and the number of iterations as 100 for all machine learning imputation algorithms. For all deep learning imputation methods, the learning rate is 0.001, the dropout rate is 0.5, the training epoch is 30, and the batch size is 128. We set the sampling size as 10 for MIWAE. In GAIN, both the generator and discriminator are modeled as a 2-layer fully connected network. NOMI is trained

Table 3: The prediction performance under MNAR

Method – Imputation	Adult			HSLs			ACS		
	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓
DRO-NOMI	0.7290	0.1662	0.3282	0.5519	0.0193	0.0617	0.6398	0.0086	<u>0.0627</u>
FairRF-NOMI	0.7383	0.1792	<u>0.2160</u>	0.5736	0.0559	0.1366	0.6237	0.0132	0.0634
SURE-NOMI	<u>0.7478</u>	<u>0.1653</u>	0.3656	<u>0.5973</u>	0.0780	0.1491	<u>0.6518</u>	0.0131	0.0764
FATE	0.7926	0.1112	0.1599	0.6578	<u>0.0557</u>	0.0711	0.7547	<u>0.0114</u>	0.0311

**Figure 2: Time consumption under different missing mechanisms****Figure 3: The prediction performance vs. missing rate R_m**

for up to 3 iterations. τ and β are set to 1.0 and 0.8 by default, respectively. In our FATE, the model training times are 150, the embedding size of tabular data is 32, the number of attention heads is 4, the dropout rate is 0.1, and the depth of the transformer is 6. To ensure a fair comparison, all baseline algorithms receive the same imputed data and each set of experiments is repeated 5 times.

6.1 Overall Performance Study

The first set of experiments studies the overall performance of fairness-intervention methods under three missing mechanisms. The experimental results under the MAR, MCAR, and MNAR mechanisms are reported in Table 1, Table 2, Table 3, respectively. For MCAR and MNAR, we only include the fairness-intervention methods using the best-performing imputation method NOMI.

We can see that in most cases, FATE substantially outperforms all baselines. Compared to the three baselines, FATE achieves an average accuracy improvement of 9.13% on *Adult*, 11.4% on *HSLs*, and 22.03% on *ACS*. Additionally, FATE remains competitive in terms of fairness, achieving an average fairness improvement of 5.69%. In particular, in terms of prediction performance, FATE exceeds the best-performing baseline (i.e., SURE with NOMI) by 9.75% on average, and it even increases up to 17.66% on the *Adult* dataset w.r.t. ACC. This is because that, FATE introduces an incomplete data embedding strategy and a missingness-aware Transformer

block for learning the observed data distribution and capturing information encoded in the missingness from incomplete tabular data, thereby enhancing the prediction accuracy. Meanwhile, the debiased Transformer block utilizes normalization and absolute value operations to reduce the discrimination risk in attention weights, which minimizes the differences between demographic groups and enhances fairness.

It is noteworthy that the prediction accuracy of FATE surpasses those of all fairness-intervention algorithms with data imputation, highlighting that relying on imputation might not be the most effective approach. Using the best accuracy baseline SURE as reference, we observe significant differences in performance among various imputation techniques. The differences in accuracy achieved by using different imputation algorithms can be as high as 5.42%, while the differences in fairness can reach up to 71.43%. This indicates that different imputation algorithms introduce varying degrees of bias, and there is no universally optimal imputation method among different datasets and missing mechanisms. We consistently utilize NOMI as the imputation algorithm for all baselines in the following experiments, due to its superior performance.

Moreover, for the MNAR mechanism, FATE significantly outperforms the baseline, especially on prediction accuracy, which is consistent with Observation 1. It is attributed to the *IDE* module of FATE, which effectively captures the complex missing state information for better representations and predictions.

Table 4: The ablation study of FATE

Datasets		Adult			HSLs			ACS		
Mechanisms	Models	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓	ACC↑	EOPP↓	EODD↓
MCAR	FATE	0.7850	0.0966	0.1536	0.6510	0.0349	0.0622	0.7526	0.0066	<u>0.0173</u>
	FATE-noM	<u>0.7793</u>	0.1523	0.4106	0.6424	0.1285	0.2044	0.7426	0.0079	0.0217
	FATE-noMML	0.7733	0.1308	<u>0.3209</u>	0.6450	0.1351	0.2064	0.7421	0.0114	0.0224
	FATE-noDAL	0.7712	<u>0.1175</u>	0.3257	<u>0.6459</u>	<u>0.1158</u>	<u>0.1752</u>	<u>0.7483</u>	<u>0.0071</u>	0.0199
MAR	FATE	0.7892	0.0944	0.1534	0.6522	0.0469	0.0672	0.7534	0.0102	0.0076
	FATE-noM	<u>0.7860</u>	0.1309	0.3384	0.6455	0.1266	0.2004	0.7438	0.0154	0.0180
	FATE-noMML	0.7863	0.1267	0.3707	<u>0.6493</u>	0.1446	0.2289	0.7461	0.0193	0.0251
	FATE-noDAL	0.7799	<u>0.1215</u>	<u>0.3296</u>	0.6487	<u>0.1109</u>	<u>0.1862</u>	<u>0.7504</u>	<u>0.13.8</u>	<u>0.0111</u>
MNAR	FATE	0.7926	0.1112	0.1599	0.6578	0.0557	0.0711	0.7549	0.0114	0.0311
	FATE-noM	<u>0.7839</u>	0.2068	<u>0.4236</u>	0.6493	0.1046	0.1709	0.7470	0.0204	0.0514
	FATE-noMML	0.7772	0.1985	0.4529	<u>0.6525</u>	0.1000	0.1583	0.7462	0.0195	0.0494
	FATE-noDAL	0.7821	<u>0.1839</u>	0.4423	0.6493	<u>0.0931</u>	<u>0.1522</u>	<u>0.7505</u>	<u>0.0192</u>	<u>0.0473</u>

Regarding the efficiency issue among methods, Figure 2 compares the time consumption of all methods under different missing mechanisms i.e., *MCAR*, *MAR*, and *MNAR*, the missing rate in this figure is 50%. One can conclude that, FATE takes the least time on model training than other fairness intervention algorithms and is more efficient. The reason is that, FATE can directly handle the incomplete tabular dataset without additional imputation costs. In contrast, every fairness-intervention baseline has to train the imputation algorithm, incurring longer training time.

6.2 Effect of Missing Rate

Figure 3 presents the experimental results of varying the missing rate R_m (i.e., the proportion of values dropped from the observed data) on the *HSLs* dataset under different missing mechanisms. We observe that FATE outperforms all baseline methods in most cases, especially regarding prediction accuracy. It further confirms the effectiveness of our proposed model FATE. As R_m increases, prediction performance consistently declines for all algorithms. It is attributed to the less observed data information when the missing rate turns high. In particular, FATE is more stable than others as R_m increases. When the missing rate varies from 10% to 80%, the reduction in prediction accuracy of FATE is, on average, 5.70%, which is lower compared to the decrease of 6.94% observed in the best-performing baseline DRO. The reason is that FATE includes the mask matrix as part of the input features and employs a missingness-aware attention mechanism to learn the data distribution from all observed data, conditional on the feature’s missing state. This approach reduces or delays the impact of an increased missing rate on prediction performance. Besides, as the missing rate increases, the fairness of the baseline algorithms gradually improves. This is because, with a higher missing rate, the performance of the imputation methods declines, resulting in more similar data distribution among sensitive groups. This leads to a marked improvement in fairness, but it comes at a substantial cost to the model’s accuracy.

6.3 Ablation Study

We examine the impact of key components in FATE on prediction accuracy and fairness. The experimental results are detailed in Table 4. FATE-noM is a variant of FATE that does not use the mask matrix $\mathbf{M}$ in the incomplete data embedding strategy. FATE-noMA

is a variant of FATE that employs a traditional self-attention mechanism instead of the missingness-aware attention in the IDE module. It employs the imputation model NOMI to estimate missing values before training. FATE-noDRL is a variant of FATE that replaces the debiased self-attention mechanism in the DRL module with a traditional self-attention mechanism. We can observe that the incomplete tabular embedding, the missingness-aware attention, and the debiased self-attention mechanism positively impact prediction performance and enhance fairness. Specifically, accuracy decreases by an average of 1.16%, 0.95%, and 0.98%, respectively. Meanwhile, fairness decreases by an average of 31.76%, 59.58%, and 35.34% in EOPP, and 58.35%, 63.52%, and 36.34% in EODD, respectively. It signifies that, the missingness-aware attention mechanism in the IDE module contributes most to FATE. It also confirms the rationality and effectiveness of the incomplete tabular embedding in IDE and debiased self-attention in DRL.

7 Conclusion

In this paper, we propose a Transformer-based fairness intervention model FATE over incomplete data. It can get rid of the bias incurred by missingness in both sensitive and non-sensitive attributes. FATE consists of two modules including incomplete data encoding (IDE) and debiased representation learning (DRL). IDE introduces an incomplete data embedding strategy and the missingness-aware attention mechanism to capture the observed data distribution and missing state information. DRL employs a debiased attention mechanism to minimize group discrimination, thereby achieving fair representation. Extensive experiments demonstrate the remarkable superiority of FATE in predication accuracy and fairness.

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